

2-1 (1)

$$\nu_0 = \frac{\phi}{h} = \frac{1.9 \times 1.6 \times 10^{-19}}{6.626 \times 10^{-34}} = 4.6 \times 10^{14} (\text{Hz})$$

$$\lambda_0 = \frac{c}{\nu_0} = \frac{3 \times 10^8}{4.6 \times 10^{14}} = 6.5 \times 10^2 (\text{nm})$$

(2)

$$\nu = \frac{E_k + \phi}{h} = \frac{(1.5 + 1.9) \times 1.6 \times 10^{-19}}{6.626 \times 10^{-34}} = 8.2 \times 10^{14} (\text{Hz})$$

$$\lambda = \frac{c}{\nu} = \frac{3 \times 10^8}{8.2 \times 10^{14}} = 3.66 \times 10^2 (\text{nm})$$

2-2 (1) 类氢原子 $\text{He}^+ (Z=2)$; $\text{Li}^{++} (Z=3)$ 的 r 、 v

$$r'_n = \frac{r_n}{Z} = \frac{n^2}{Z} r_1 = 0.0529 \frac{n^2}{Z} (\text{nm})$$

$$v'_n = Z v_n = \frac{Z}{n} v_1 = 2.2 \times 10^6 \frac{Z}{n} (\text{m/s})$$

(2) $\text{He}^+ (Z=2)$ 、 $\text{Li}^{++} (Z=3)$ 在基态 ($n_b=1$) 的结合能:

$$E = E_\infty - E_{k_b} = -E_{k_b} = 13.6 \frac{Z^2}{n_b^2} (\text{eV})$$

(3) 从基态到第一激发态的激发能:

$$\Delta E = E_2 - E_1 = 13.6 \left(1 - \frac{1}{2^2}\right) Z^2 (\text{eV})$$

从第一激发态到基态激发的光:

$$\lambda'_{nk} = \frac{1}{Z^2} \lambda_{21} = \frac{1}{Z^2} \left[R_H \left(\frac{1}{1^2} - \frac{1}{2^2} \right) \right]^{-1}$$

$$2-3 \quad E'_n = -13.6 \frac{Z^2}{n^2} (\text{eV})$$

$$E_k = E'_2 - E'_1 = -13.6 \times 3^2 \times \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = 91.8 (\text{eV})$$

$$2-7 \quad \lambda'_{nk} = \frac{1}{Z^2} \lambda_{nk}$$

$$\lambda'_{32} - \lambda'_{21} = 133.7 (\text{nm})$$

$$\frac{1}{Z^2} (\lambda_{32} - \lambda_{21}) = 133.7 (\text{nm})$$

$$Z^2 = \frac{(\lambda_{32} - \lambda_{21})}{133.7} = \frac{121.6 - 656.3}{133.7} = 4$$

$$\therefore Z = 2$$

2-8

$$Z = 2$$

$$E'_n = -13.6 \frac{Z^2}{n^2} (\text{eV})$$

$$E'_{21} = E'_2 - E'_1 = -13.6 \times 2^2 \times \left(\frac{1}{2^2} - \frac{1}{1^2} \right) = 40.8 (\text{eV})$$

$$\frac{1}{2} m v^2 = E'_{21} - 13.6 = 27.2 (\text{eV})$$

$$v = \sqrt{\frac{2 \times 27.2 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}} = 3.09 \times 10^6 (\text{m/s})$$

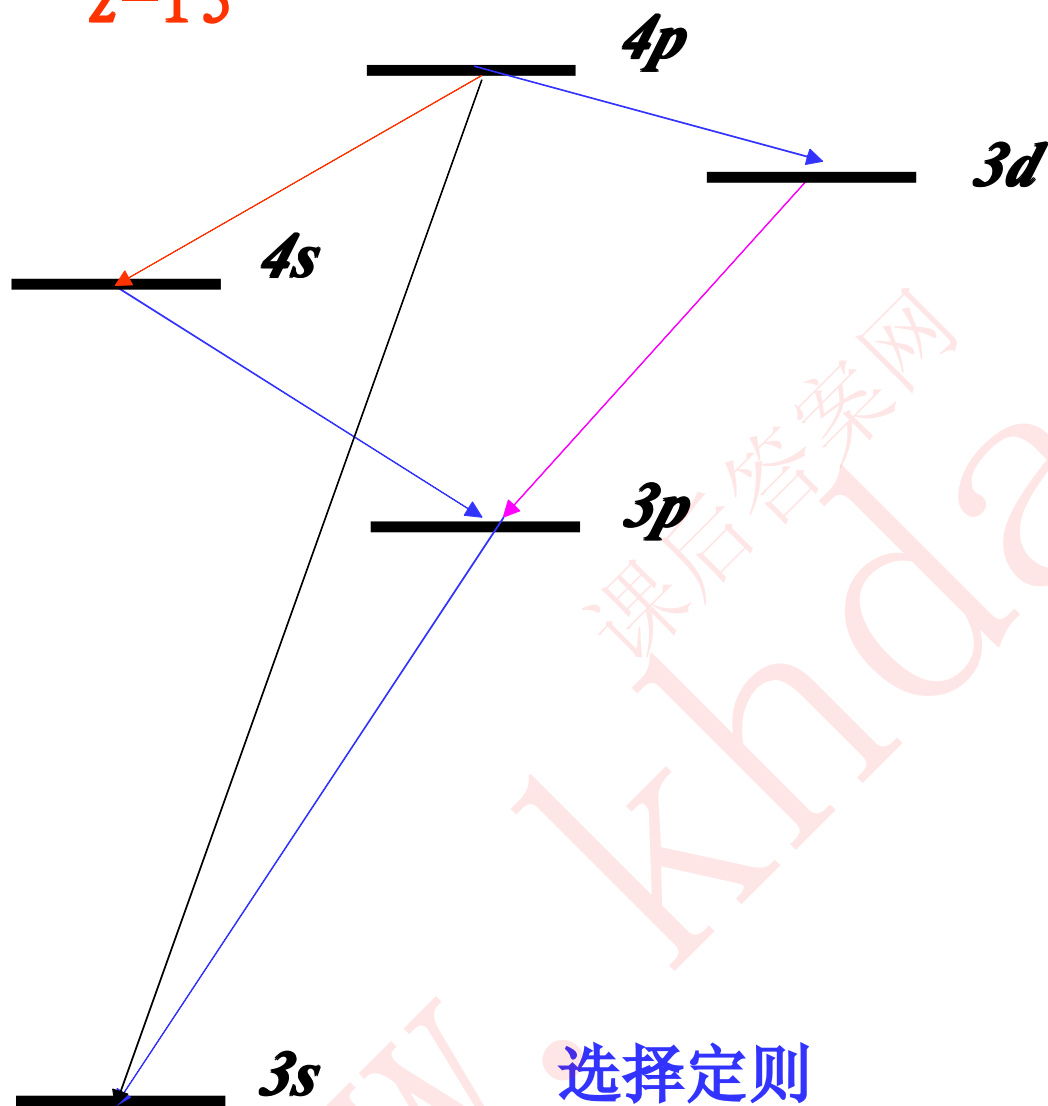
2-11

$$\mu_H = \frac{m_H m_e}{m_H + m_e} \quad \mu_D = \frac{m_D m_e}{m_D + m_e}$$

$$\therefore \frac{R_H}{R_D} = \frac{\mu_H}{\mu_D} = \frac{\frac{m_H m_e}{m_H + m_e}}{\frac{m_D m_e}{m_D + m_e}} = \frac{1 + \frac{m_e}{m_D}}{1 + \frac{m_e}{m_H}} = \frac{1 + \frac{m_H}{m_D} \cdot \frac{m_e}{m_H}}{1 + \frac{m_e}{m_H}}$$

$$\therefore \frac{m_p}{m_e} = \frac{m_H}{m_e} = \frac{\frac{R_D}{R_H} - \frac{m_D}{m_H}}{1 - \frac{R_D}{R_H}} = \frac{0.999728 - 0.50020}{1 - 0.999728} = 1836$$

2-13



选择定则

$$\Delta l = \pm 1$$

$$\frac{1}{\lambda_1} = T(4s) - T(4p)$$

$$\frac{1}{\lambda_2} = T(3s) - T(4p)$$

$$\frac{1}{\lambda_3} = T(3d) - T(4p)$$

$$\frac{1}{\lambda_4} = T(3p) - T(4s)$$

$$\frac{1}{\lambda_5} = T(3p) - T(3d)$$

$$\frac{1}{\lambda_6} = T(3s) - T(3p)$$

$$2-14 (1) \because \frac{1}{\lambda_{\infty}} = T(3p) - T(\infty) = T(3p)$$

$$\therefore T(3p) = \frac{1}{\lambda_{\infty}} = \frac{1}{408.6 \times 10^{-9}} = 2.447 \times 10^6 (\text{m}^{-1})$$

钠原子的共振线波长: $\lambda_c = \lambda_{3p \rightarrow 3s}$

$$\therefore \frac{1}{\lambda_c} = T(3s) - T(3p)$$

$$\therefore T(3s) = T(3p) + \frac{1}{\lambda_c} = 2.447 \times 10^6 + \frac{1}{589.3 \times 10^{-9}} = 4.144 \times 10^6 (\text{m}^{-1})$$

$$\therefore E = -hcT$$

$$\therefore E(3p) = -hcT(3p) = -3.03 (\text{eV})$$

$$E(3s) = -hcT(3s) = -5.14 (\text{eV})$$

(2) 钠原子基态电子的电离能: $E_i = -E(3s) = 5.14(\text{eV})$

从基态到第一激发态的激发能:

$$\Delta E = E(3p) - E(3s) = -3.03 - (-5.14) = 2.11(\text{eV})$$

$$3-1 \quad \because p = \sqrt{2mE_k} = \frac{h}{\lambda}$$

$$\therefore \lambda = \frac{h}{\sqrt{2mE_k}} = \frac{1.226}{\sqrt{E_k(\text{eV})}} = \begin{cases} 0.39 \text{ (nm)} (E_k = 10 \text{ eV}) \\ 0.123 \text{ (nm)} (E_k = 100 \text{ eV}) \\ 0.039 \text{ (nm)} (E_k = 1000 \text{ eV}) \end{cases}$$

$$3-2 (1) \quad \because p = \frac{h}{\lambda}; \lambda_e = \lambda_\phi = 0.4 \text{ (nm)}$$

$$\therefore \frac{p_\phi}{p_e} = 1$$

$$(2) \quad \because \varepsilon = h\nu = \frac{hc}{\lambda} = \frac{1240}{0.4} = 3100 \text{ (eV)}$$

$$E_k = \frac{p^2}{2m_e} = \frac{h^2}{2m_e \lambda_e^2} = \left(\frac{1.226}{\lambda_e} \right)^2 = \left(\frac{1.226}{0.4} \right)^2 = 9.4 \text{ (eV)}$$

$$\therefore \frac{\varepsilon}{E_k} = \frac{3100}{9.4} \approx 3300$$

$$3-3 \quad \because E_k = m_e c^2 - m_0 c^2 = m_0 c^2$$

$$\therefore m_e = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = 2m_0$$

$$\Rightarrow v = \frac{\sqrt{3}}{2} c$$

$$\begin{aligned} \therefore \lambda &= \frac{h}{m_e v} \\ &= \frac{h}{2m_0 v} = \frac{h}{\sqrt{3} m_0 c} \\ &\approx 0.0014(\text{nm}) \end{aligned}$$

3-4

设 a 为晶格常数, d 为晶面间距

$$2d \sin \alpha = k\lambda$$

$$k = 1, 2, 3, \dots$$

$$a \sin \theta = k\lambda$$

$$k = 1, 2, 3, \dots$$

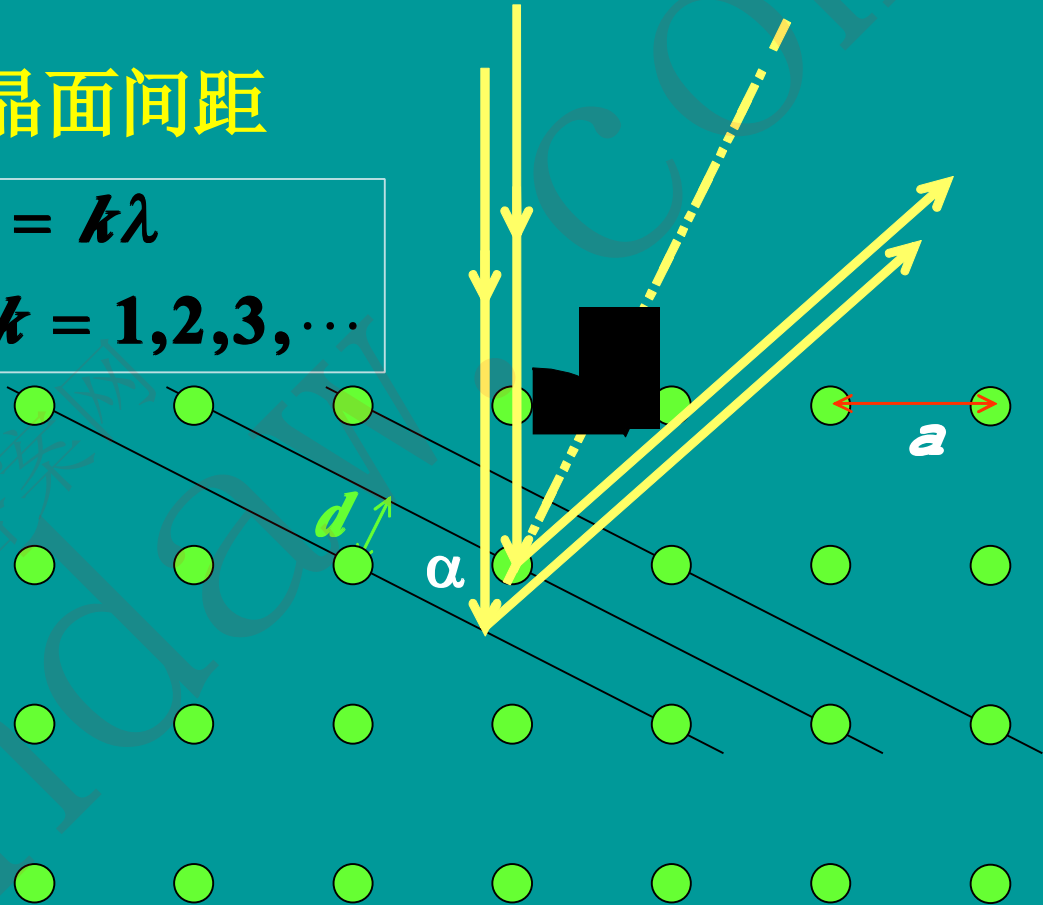
$$\because d = 0.18 \text{ nm}, \alpha = 30^\circ, k = 1$$

$$\therefore \lambda = 2 \times d \sin \alpha$$

$$= 2 \times 0.18 \times \sin 30^\circ = 0.18 \text{ nm}$$

$$\begin{aligned} \therefore p &= \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{0.18 \times 10^{-9}} \\ &= 3.68 \times 10^{-24} \text{ kg} \cdot \text{m} / \text{s} \end{aligned}$$

$$\begin{aligned} \therefore E_k &= \frac{p^2}{2m} = \frac{(3.68 \times 10^{-24})^2}{2 \times 1.67 \times 10^{-27}} \\ &= 4.06 \times 10^{-21} \text{ (J)} = \frac{4.06 \times 10^{-21}}{1.60 \times 10^{-19}} = 0.025 \text{ (eV)} \end{aligned}$$



3-7

$$\because E = h\nu = \frac{hc}{\lambda}; dE = -\frac{hc}{\lambda^2} d\lambda$$

$$\therefore \Delta E \approx \frac{hc}{\lambda^2} \Delta \lambda$$

$$\because \Delta E \cdot \Delta t \geq \frac{\hbar}{2} = \frac{h}{4\pi}$$

$$\begin{aligned} \therefore \tau = \Delta t &\geq \frac{h}{4\pi \Delta E} \\ &= \frac{\lambda}{4\pi c \cdot \frac{\Delta \lambda}{\lambda}} = \frac{600 \times 10^{-9}}{4\pi c \times 10^{-7}} \\ &\approx 2 \times 10^{-9} (s) \end{aligned}$$

3-8 电子束缚在10 fm 线度(原子核线度的数量级), 试用不确定度关系估算电子的最小动能。 $\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2}$

解: 电子位置的不确定量 $\Delta x=10^{-14}\text{m}$, 由不确定度关系得

$$\Delta p_x \geq \frac{\hbar}{2\Delta x} = \frac{1.05 \times 10^{-34}}{2 \times 10^{-14}} = 0.53 \times 10^{-20} \text{ kg} \cdot \text{m/s}$$

取最小动量为: $p = 0.53 \times 10^{-20} \text{ kg} \cdot \text{m/s}$

$$\therefore v = \frac{p}{m} = \frac{0.53 \times 10^{-20} \text{ kg} \cdot \text{m/s}}{9.11 \times 10^{-31} \text{ kg}} > c(3 \times 10^8 \text{ m/s})$$

所以需要应用相对论的能量动量公式

$$E = \sqrt{p^2 c^2 + m_0^2 c^4} \approx pc \approx 1.6 \times 10^{-12} \text{ J} \approx 10 \text{ MeV}$$

电子的动能

$$E_k = E - m_0 c^2 \approx 1.6 \times 10^{-12} \text{ J} = 10 \text{ MeV}$$

讨论: 电子具有这样大的动能足以把原子核击碎从而判断原子核由质子和电子组成是不可能的。

3-9 (1)

$$\psi = N e^{-\frac{|x|}{2a} - \frac{|y|}{2b} - \frac{|z|}{2c}}$$

$$\therefore \int_{-\infty}^{\infty} \psi^* \psi d\vec{r} = N^2 \int_{-\infty}^{\infty} e^{-\frac{|x|}{a} - \frac{|y|}{b} - \frac{|z|}{c}} dx \cdot dy \cdot dz$$

$$= 8 N^2 \int_0^{\infty} e^{-\frac{x}{a}} dx \cdot \int_0^{\infty} e^{-\frac{y}{b}} dy \cdot \int_0^{\infty} e^{-\frac{z}{c}} dz$$

$$\therefore N = \frac{1}{\sqrt{8abc}} = 8 N^2 abc = 1$$

$$(2) P = N^2 \int_0^a e^{-\frac{x}{a}} dx \cdot \int_{-\infty}^{\infty} e^{-\frac{y}{b}} dy \cdot \int_{-\infty}^{\infty} e^{-\frac{z}{c}} dz$$

$$= 4 N^2 bc \cdot \int_0^a e^{-\frac{x}{a}} dx = \frac{1}{2} \left(1 - \frac{1}{e} \right)$$

$$(3) P = N^2 \int_{-\infty}^{\infty} e^{-\frac{x}{a}} dx \cdot \int_{-b}^b e^{-\frac{y}{b}} dy \cdot \int_{-c}^c e^{-\frac{z}{c}} dz$$

$$= 8 N^2 a \cdot \int_0^b e^{-\frac{y}{b}} dy \cdot \int_0^c e^{-\frac{z}{c}} dz = \left(1 - \frac{1}{e} \right)^2$$

$$\text{3-11} \quad \psi_n(x) = \begin{cases} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right); & n = 1, 2, 3, \dots; 0 \leq x \leq a \\ 0; & x < a, x > a \end{cases}$$

$$\bar{x} = \int_{-\infty}^{\infty} \psi_0^* x \psi_0 dx = \frac{2}{a} \int_0^a x \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a}{2}$$

$$\bar{x}^2 = \int_{-\infty}^{\infty} \psi_0^* x^2 \psi_0 dx = \frac{2}{a} \int_0^a x^2 \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{a^2}{3} - \frac{a^2}{2n^2\pi^2}$$

$$\overline{(x - \bar{x})^2} = \bar{x}^2 - \bar{x}^2 = \frac{a^2}{12} - \frac{a^2}{2n^2\pi^2}$$

$$(2) \quad n \rightarrow \infty, \overline{(x - \bar{x})^2} = \frac{a^2}{12}$$

经典:

$$\bar{x} = \int_0^a x \frac{1}{a} dx = \frac{a}{2} \quad \bar{x}^2 = \int_0^a x^2 \frac{1}{a} dx = \frac{a^2}{3}$$

$$\overline{(x - \bar{x})^2} = \bar{x}^2 - \bar{x}^2 = \frac{a^2}{12}$$

3-12

(1) 氢原子1S态波函数:

$$\psi_{1,0,0}(r) = \frac{1}{\sqrt{\pi a_1^3}} e^{-\frac{r}{a_1}}$$

$r \rightarrow r+dr$ 的几率:

$$w(r)dr = |\psi_{1,0,0}(r)|^2 4\pi r^2 dr = \frac{4}{a_1^3} r^2 e^{-\frac{2r}{a_1}} dr$$

电荷密度:

$$\rho(r) = -e w(r) = -e \frac{4}{a_1^3} r^2 e^{-\frac{2r}{a_1}}$$

电荷密度取极大值条件:

$$\frac{d\rho(r)}{dr} = -e \frac{4}{a_1^3} \frac{d}{dr} \left(r^2 e^{-\frac{2r}{a_1}} \right) = 0$$

$$\Rightarrow r = a_1 = 0.0529 \text{ nm}, \rho = \rho_{\max}$$

(2) 氢原子2P态波函数:

$$\psi_{2,1,0}(r) = \frac{r}{4a_1\sqrt{2\pi a_1^3}} e^{-\frac{r}{2a_1}} \cos\theta$$

$$w(r)dr = \int_0^\pi d\theta \int_0^{2\pi} d\varphi |\psi_{2,1,0}(r)|^2 r^2 dr$$

$$w(r) = \frac{r^4}{32\pi a_1^5} e^{-\frac{r}{a_1}} \int_0^\pi \cos^2\theta \sin\theta d\theta \int_0^{2\pi} d\varphi$$

$$= \frac{1}{24a_1^5} r^4 e^{-\frac{r}{a_1}}$$

$$\rho(r) = -e w(r) = -\frac{e}{24a_1^5} r^4 e^{-\frac{r}{a_1}}$$

$$\frac{d\rho(r)}{dr} = -e \frac{1}{24a_1^5} \frac{d}{dr} \left(r^4 e^{-\frac{r}{a_1}} \right) = 0$$

$$\Rightarrow r = 4a_1 = 0.212nm, \rho = \rho_{\max}$$

3-13

氢原子基态波函数:

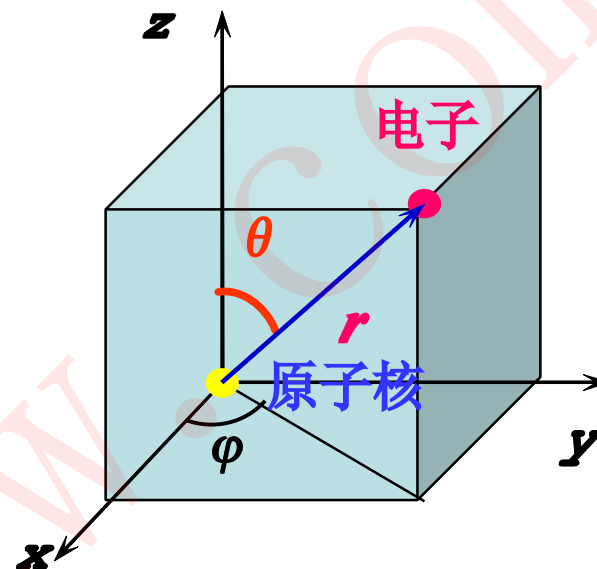
$$\psi_{1,0,0}(\mathbf{r}) = \frac{1}{\sqrt{\pi a_1^3}} e^{-\frac{r}{a_1}}$$

$$\hat{U}(\mathbf{r}) = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

$$\langle U \rangle = \int_{-\infty}^{\infty} \psi_{1,0,0}^* \hat{U}(\mathbf{r}) \psi_{1,0,0} d\mathbf{r}$$

$$= \int_0^{\infty} |\psi_{1,0,0}(\mathbf{r})|^2 \left(\frac{-e^2}{4\pi\epsilon_0 r} \right) 4\pi r^2 dr$$

$$= \int_0^{\infty} \frac{-e^2}{\pi\epsilon_0 a_1^3} r e^{-\frac{2r}{a_1}} dr = -\frac{1}{4\pi\epsilon_0} \frac{e^2}{a_1}$$



4-1

$$\boldsymbol{E} = \mu_{zs} \cdot \boldsymbol{B} = g_s m_s \mu_B \boldsymbol{B}$$

$$\because g_s = 2, m_s = \pm \frac{1}{2}$$

$$\therefore \boldsymbol{E} = \begin{cases} \mu_B \boldsymbol{B} \\ -\mu_B \boldsymbol{B} \end{cases}$$

$$\begin{aligned} \Delta E &= 2\mu_B B = 2 \times 0.5788 \times 10^{-4} \times 1.2 \\ &= 1.39 \times 10^{-4} (eV) \end{aligned}$$

4-2 ${}^2D_{3/2} : l=2; j=3/2; 2s+1=2, S=1/2$

朗德因子

$$g = \frac{3}{2} + \frac{s(s+1) - l(l+1)}{2j(j+1)} = \frac{3}{2} + \frac{\frac{1}{2} \times \frac{3}{2} - 2 \times 3}{2 \times \frac{3}{2} \times \frac{5}{2}} = \frac{4}{5}$$

$$\mu_j = g_j \sqrt{j(j+1)} \mu_B = \frac{4}{5} \sqrt{\frac{3}{2} \times \frac{5}{2}} \mu_B = 1.55 \mu_B$$

$$\mu_{jz} = -g_j m_j \mu_B = \begin{cases} \frac{6}{5} \mu_B & m_j = \frac{3}{2} \\ \frac{2}{5} \mu_B & m_j = \frac{1}{2} \\ -\frac{2}{5} \mu_B & m_j = -\frac{1}{2} \\ -\frac{6}{5} \mu_B & m_j = -\frac{3}{2} \end{cases}$$

玻尔磁子

4-3 ${}^6G_{3/2} : l=4; j=3/2; 2s+1=6, S=5/2$

$$g = 1 + \frac{j(j+1) - l(l+1) + s(s+1)}{2j(j+1)}$$

$$= \frac{3}{2} + \frac{s(s+1) - l(l+1)}{2j(j+1)}$$

$$= \frac{3}{2} + \frac{\frac{5}{2} \times \frac{7}{2} - 4 \times 5}{2 \times \frac{3}{2} \times \frac{5}{2}} = 0$$

$$\hat{\mu} = -g \frac{e}{2m} \hat{j} = 0$$

4-4 $^2S_{1/2} : l=0; j=1/2; 2s+l=2, S=1/2$

朗德因子

$$g = \frac{3}{2} + \frac{s(s+1) - l(l+1)}{2j(j+1)} = \frac{3}{2} + \frac{\frac{1}{2} \times \frac{3}{2} - 0}{2 \times \frac{1}{2} \times \frac{3}{2}} = 2$$

$$\therefore Z = M_j g \mu_B \frac{\partial B}{\partial z} \frac{d \cdot D}{2 E_K}$$

$$M_j = \frac{1}{2} \quad \mu_B \equiv \frac{e\hbar}{2m} = 0.927 \times 10^{-23} \text{ (J/T)}$$

$$\therefore \frac{\partial B}{\partial z} = \frac{m v^2 \cdot Z}{M_j g \mu_B d \cdot D}$$

$$\begin{aligned} &= \frac{107.87 \times 1.6 \times 10^{-27} \times 400^2 \times 1 \times 10^{-3}}{0.927 \times 10^{-23} \times 10 \times 10^{-2} \times 25 \times 10^{-2}} \\ &= 1.2 \times 10^2 \left(\frac{T}{m} \right) \end{aligned}$$

4-11

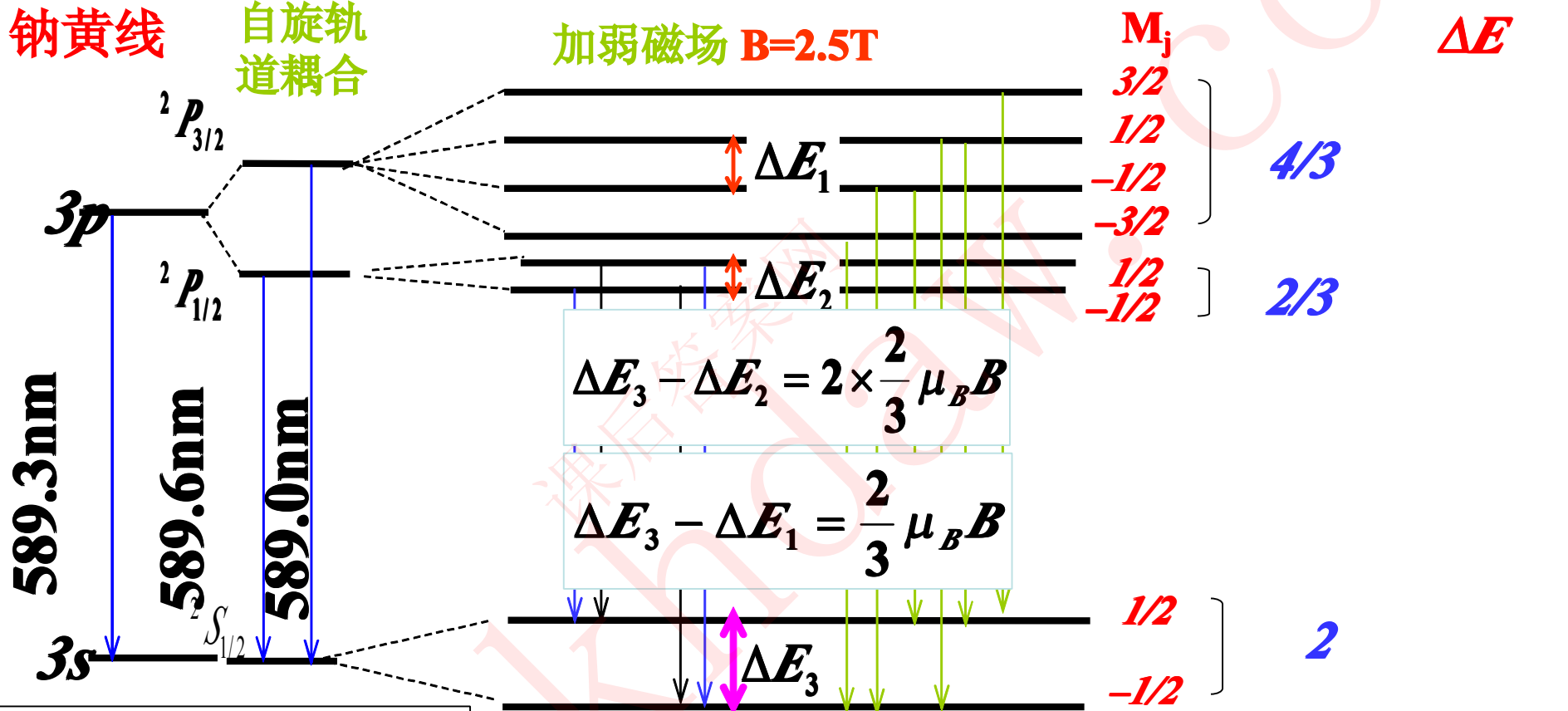
$$g = \frac{3}{2} + \frac{s(s+1) - l(l+1)}{2j(j+1)}$$

$$\Delta E = E_{jB} \mid_{\Delta m=1} = g\mu_B B$$

钠黄线

自旋轨道耦合

加弱磁场 $B=2.5T$



$$\mu_B \boldsymbol{B} \equiv \hbar \omega_L = \hbar \nu_L$$

$$\Delta \tilde{\nu}_L = \frac{\mu_B B}{hc} = 46.7 B [SI]$$

$$\Delta \tilde{v} = \frac{\Delta E}{\mu_B B} \times 46.7 B [SI]$$



$\Delta v_{D1 \rightarrow D2} = \frac{1}{\lambda_2} - \frac{1}{\lambda_1} = \frac{1}{589.0} - \frac{1}{589.6} = 1.73 \times 10^{-6} \text{ nm}^{-1}$

$\Delta \tilde{v}_1 = \frac{2}{3} \times 46.7 \times 2.5 = 78 [\text{m}^{-1}]$

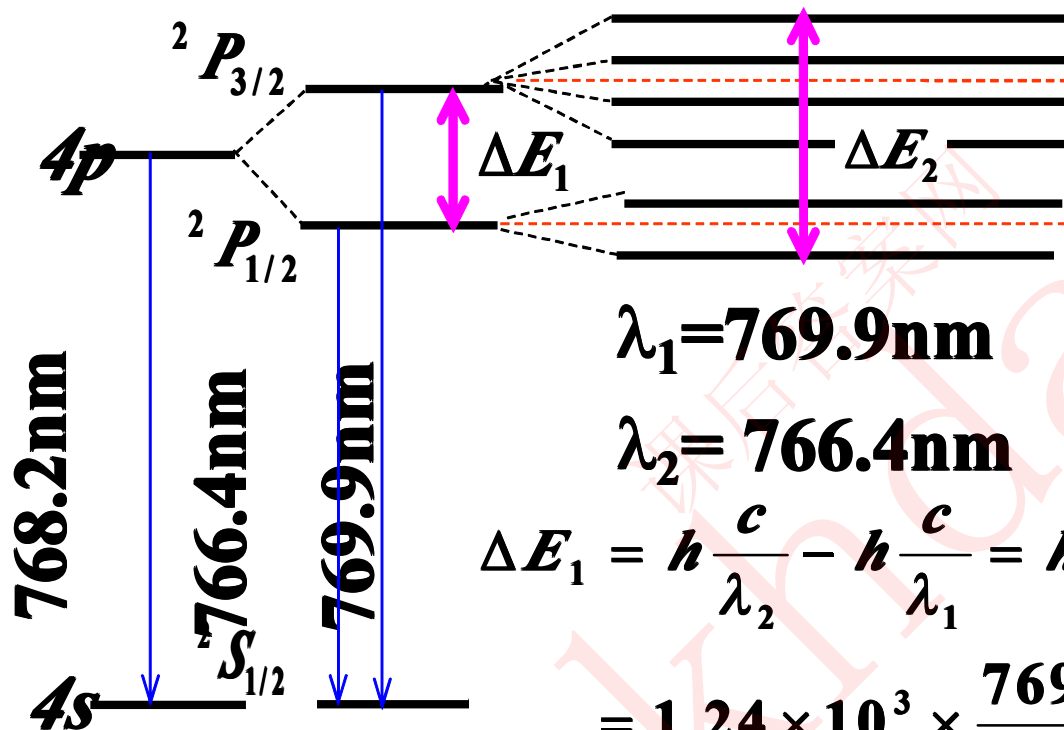
4-12

钾K

自旋轨道耦合

加弱磁场

$$g = \frac{3}{2} + \frac{s(s+1) - l(l+1)}{2j(j+1)}$$



M_j

ΔE

$3/2$

$1/2$

$-1/2$

$-3/2$

$1/2$

$-1/2$

4/3

2/3

$$\lambda_1 = 769.9 \text{ nm}$$

$$\lambda_2 = 766.4 \text{ nm}$$

$$\Delta E = E_{jB} |_{\Delta m=1} = g \mu_B B$$

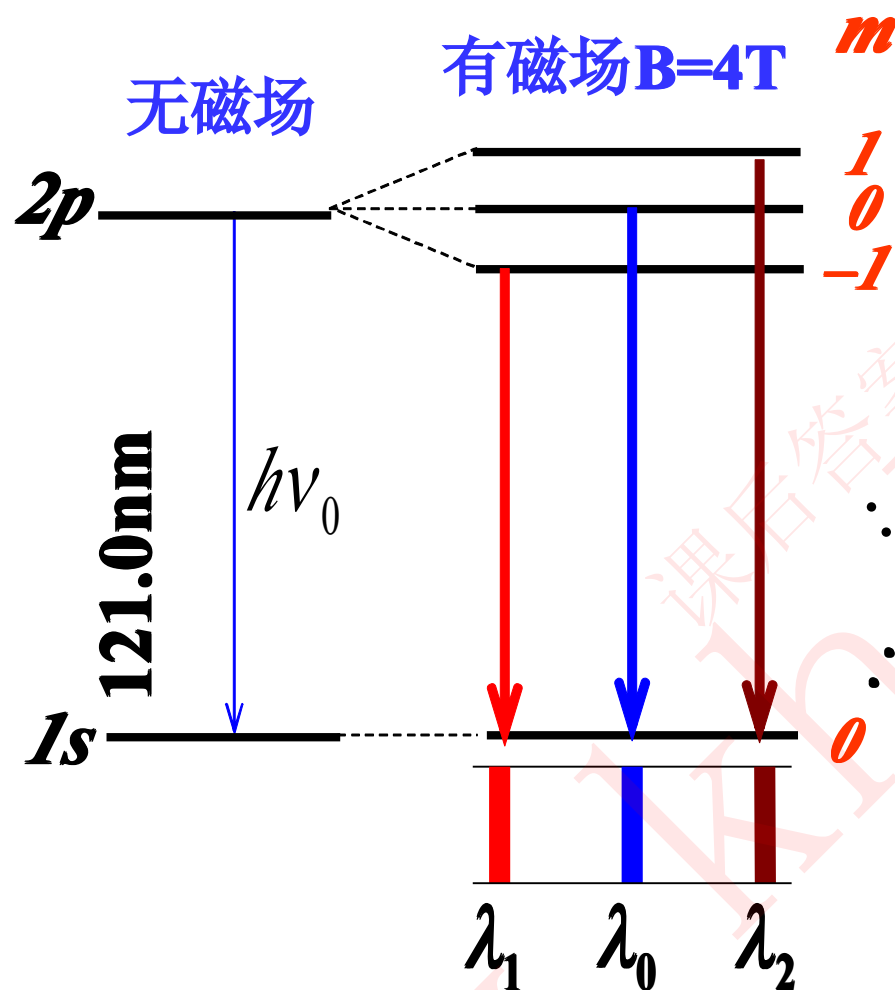
$$\Delta E_1 = h \frac{c}{\lambda_2} - h \frac{c}{\lambda_1} = hc \frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2}$$

$$= 1.24 \times 10^3 \times \frac{769.9 - 766.4}{769.9 \times 766.4} = 7.35 \times 10^{-3} \text{ eV}$$

$$\therefore \Delta E_2 = \Delta E_1 + 1.5 \times \frac{4}{3} \times \mu_B B + 0.5 \times \frac{2}{3} \times \mu_B B = 1.5 \Delta E_1$$

$$\therefore B = \frac{3}{14 \mu_B} \Delta E_1 = \frac{3}{14 \times 0.5788 \times 10^{-4}} \times 7.35 \times 10^{-3} = 27.2 (T)$$

4-14. H 原子及在强磁场中 $2P \rightarrow 1S$ 的塞曼效应



$$\Delta E = \mu_B B = h\nu_L$$

$$\Delta E = h \frac{c}{\lambda_2} - h \frac{c}{\lambda_1} = hc \Delta \tilde{\nu}$$

$$\Delta \tilde{\nu}_L = \frac{\mu_B B}{hc} = 46.7 \times 4 = 187 [\text{m}^{-1}]$$

$$\because \lambda = \frac{1}{\tilde{\nu}} \quad \lambda_0 = 121.0 \text{ nm}$$

$$\because \Delta \lambda = -\frac{1}{\tilde{\nu}^2} \Delta \tilde{\nu} = -\lambda_0^2 \Delta \tilde{\nu}$$

$$= -(121.0)^2 \times 187 \times 10^{-9}$$

$$= -2.74 \times 10^{-3} \text{ nm}$$

$$\therefore \lambda_2 = \lambda_0 - |\Delta \lambda| = 121.0 - 2.74 \times 10^{-3} \text{ nm}$$

$$\lambda_1 = \lambda_0 + |\Delta \lambda| = 121.0 + 2.74 \times 10^{-3} \text{ nm}$$

5-1.

氮原子中电子的结合能为: **24.5 eV**

电离氮原子中一个电子须提供能量为: **24.5 eV**

电离了一个电子的氮离子为类氢原子

用玻尔的方法可以求出类氢原子的能量

$$E_n = Z^2 E_n(H) = -Z^2 \frac{13.6}{n^2} (eV)$$

$$\because Z = 2$$

$$\therefore E'_1 = -2^2 \times 13.6 (eV) = -54.4 (eV)$$

即电离氮原子中第二个电子须提供能量为: **54.4 eV**

所以电离氮原子中第二个电子须提供能量为:

$$24.5 \text{ eV} + 54.4 \text{ eV} = 78.9 \text{ eV}$$

5-2.

$$^4D_{3/2} \Rightarrow S = \frac{3}{2}, L = 2, J = \frac{3}{2}$$

$$\therefore \hat{\vec{J}} = \hat{\vec{L}} + \hat{\vec{S}}$$

$$\therefore \hat{\vec{J}}^2 = (\hat{\vec{L}} + \hat{\vec{S}})^2 = \hat{\vec{L}}^2 + 2\hat{\vec{L}} \cdot \hat{\vec{S}} + \hat{\vec{S}}^2$$

$$\hat{\vec{L}} \cdot \hat{\vec{S}} = \frac{1}{2} (\hat{\vec{J}}^2 - \hat{\vec{L}}^2 - \hat{\vec{S}}^2)$$

$$\vec{L} \cdot \vec{S} = \frac{\hbar^2}{2} [J(J+1) - L(L+1) - S(S+1)]$$

$$= \frac{\hbar^2}{2} \left[\frac{3}{2} \left(\frac{3}{2} + 1 \right) - 2(2+1) - \frac{3}{2} \left(\frac{3}{2} + 1 \right) \right]$$

$$= -3\hbar^2$$

$$5-4. \because \hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$$

$$\therefore \hat{\mathbf{S}} = \hat{\mathbf{J}} - \hat{\mathbf{L}}$$

$$\hat{\mathbf{S}}^2 = \left(\hat{\mathbf{J}} - \hat{\mathbf{L}} \right)^2 = \hat{\mathbf{J}}^2 - 2\hat{\mathbf{L}} \cdot \hat{\mathbf{J}} + \hat{\mathbf{L}}^2$$

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{J}} = \frac{1}{2} \left(\hat{\mathbf{J}}^2 - \hat{\mathbf{S}}^2 + \hat{\mathbf{L}}^2 \right)$$

$$|\vec{\mathbf{L}}| \cdot |\vec{\mathbf{J}}| \cos(\vec{\mathbf{L}}, \vec{\mathbf{J}}) = \frac{1}{2} (\vec{\mathbf{J}}^2 - \vec{\mathbf{S}}^2 + \vec{\mathbf{L}}^2)$$

$$\cos(\vec{\mathbf{L}}, \vec{\mathbf{J}}) = \frac{\vec{\mathbf{J}}^2 + \vec{\mathbf{L}}^2 - \vec{\mathbf{S}}^2}{2 |\vec{\mathbf{L}}| |\vec{\mathbf{J}}|} = \frac{J(J+1) + L(L+1) - S(S+1)}{2 \sqrt{J(J+1)} \cdot \sqrt{L(L+1)}}$$

$$^3F_2 \Rightarrow S=1, L=3, J=2$$

$$\cos(\vec{\mathbf{L}}, \vec{\mathbf{J}}) = \frac{3 \times 4 + 2 \times 3 - 1 \times 2}{2 \sqrt{3 \times 4} \cdot \sqrt{2 \times 3}} = 0.9428 \quad \angle(\vec{\mathbf{L}}, \vec{\mathbf{J}}) = 19^\circ 28'$$

5-7. (1) $np^4 \Leftrightarrow np^2 \quad np^2 (l=1; m_l = 1, 0, -1)$

量子态 序号	$(m_l, m_s)_1$	$(m_l, m_s)_2$	$(m_l, m_s)_1 (m_l, m_s)_2$
1	(1, +)	(1, +)	12; 13; 14; 15; 16
2	(1, -)	(1, -)	23; 24; 25; 26
3	(0, +)	(0, +)	34; 35; 36
4	(0, -)	(0, -)	45; 46
5	(-1, +)	(-1, +)	56
6	(-1, -)	(-1, -)	

有效的量子态个数: $C_6^2 = \frac{6 \times 5}{2} = 15$

np^2 的态项: $^1S_0; ^3P_{2,1,0}; ^1D_2$

(1) $np^4 \Leftrightarrow np^2 \quad np^4 \Rightarrow ^1S_0; ^3P_{2,1,0}; ^1D_2 \Rightarrow ^3P_2$

(2) $np^5 \Leftrightarrow np^1 \quad np^1 (L=1; S=1/2) \Rightarrow ^2P_{3/2,1/2} \quad np^5 \Rightarrow ^2P_{3/2,1/2} \Rightarrow ^2P_{3/2}$

(3) $n_1 dn_2 d(l_1 = 2; l_2 = 2)$ LS耦合 $\hat{J} = \hat{L} + \hat{S}$

总自旋 $\hat{S} = \hat{s}_1 + \hat{s}_2$

$$S = \begin{cases} s_1 + s_2 \\ |s_1 - s_2| \end{cases} = \begin{cases} \frac{1}{2} + \frac{1}{2} = 1; 2S+1=3 \\ \left| \frac{1}{2} - \frac{1}{2} \right| = 0; 2S+1=1 \end{cases}$$

总角动量 $\hat{L} = \hat{l}_1 + \hat{l}_2$

$$L = \begin{cases} l_1 + l_2 \\ \dots \\ |l_1 - l_2| \end{cases} = \begin{cases} 2-2=0; \Rightarrow {}^1S \quad {}^3S \\ 0+1=1; \Rightarrow {}^1P \quad {}^3P \\ 0+2=2; \Rightarrow {}^1D \quad {}^3D \\ 0+3=3; \Rightarrow {}^1F \quad {}^3F \\ 2+2=4; \Rightarrow {}^1G \quad {}^3G \end{cases}$$

$$J = \begin{cases} L + S \\ \dots \\ |L - S| \end{cases} = \begin{cases} 0; L=0, S=0 \Rightarrow {}^1S_0 \\ 1; L=0, S=1 \Rightarrow {}^3S_1 \\ 1; L=1, S=0 \Rightarrow {}^1P_1 \\ 2, 1, 0; L=1, S=1 \Rightarrow {}^3P_{2,1,0} \\ 2; L=2, S=0 \Rightarrow {}^1D_2 \\ 3, 2, 1; L=2, S=1 \Rightarrow {}^3D_{3,2,1} \\ 3; L=3, S=0 \Rightarrow {}^1F_3 \\ 4, 3, 2; L=3, S=1 \Rightarrow {}^3F_{4,3,2} \\ 4; L=4, S=0 \Rightarrow {}^1G_4 \\ 5, 4, 3; L=4, S=1 \Rightarrow {}^3G_{5,4,3} \end{cases}$$

5-11.

氦原子基态 ${}_2\text{He}$: $1s^2$

$$1s^2 (L=0; S=0,1) \Rightarrow {}^1S_0 \Rightarrow J=0 \Rightarrow 2J+1=1$$

在施忒恩盖拉赫实验中，基态氦原子将形成 1 束原子射线束。

硼原子基态 ${}_5\text{B}$: $1s^2 2s^2 2p$

$$\begin{aligned} {}^1_2p^1 (L=1; S=1/2) &\Rightarrow {}^2P_{3/2,1/2} \Rightarrow {}^2P_{1/2} \\ &\Rightarrow J=1/2 \Rightarrow 2J+1=2 \end{aligned}$$

在施忒恩盖拉赫实验中，基态硼原子将分裂成 2 束原子射线束。

5-12.

磷原子基态 $_{15}\text{P}$: $1s^2 2s^2 2p^6 3s^2 3p^3$

$$3p^3 \Rightarrow {}^4S_{3/2}; {}^2P_{3/2,1/2}; {}^2D_{5/2,3/2} \Rightarrow {}^4S_{3/2}$$

硫原子基态 $_{16}\text{S}$: $1s^2 2s^2 2p^6 3s^2 3p^4$

$$3p^4 \Rightarrow {}^1S_0; {}^3P_{2,1,0}; {}^1D_2 \Rightarrow {}^3P_2$$

氯原子基态 $_{17}\text{Cl}$: $1s^2 2s^2 2p^6 3s^2 3p^5$

$$3p^5 \Rightarrow {}^2P_{3/2,1/2} \Rightarrow {}^2P_{3/2}$$

氩原子基态 $_{18}\text{Ar}$: $1s^2 2s^2 2p^6 3s^2 3p^6$

$$3p^6 \Rightarrow L = S = 0 \Rightarrow {}^1S_0$$

$$6-1 \quad \because \lambda_{\min} = \frac{hc}{e \cdot V} = \frac{1.24}{V[\text{kV}]} (\text{nm})$$

$$\therefore V = \frac{1.24}{\lambda_{\min} [\text{nm}]} = \frac{1.24}{0.0124} = 100 (\text{kV})$$

$$6-2 \quad \because \frac{1}{\lambda_{K\alpha}} = R \left(1 - \frac{1}{4} \right) (Z - 1)^2 = \frac{3}{4} R (Z - 1)^2$$

$$\therefore Z = \sqrt{\frac{4}{3 R \lambda_{K\alpha}}} + 1 \approx \sqrt{\frac{121.5 [\text{nm}]}{\lambda_{K\alpha}}} + 1$$

$$= \sqrt{\frac{121.5 [\text{nm}]}{0.0685}} + 1 = 43 \quad \text{Tc}$$

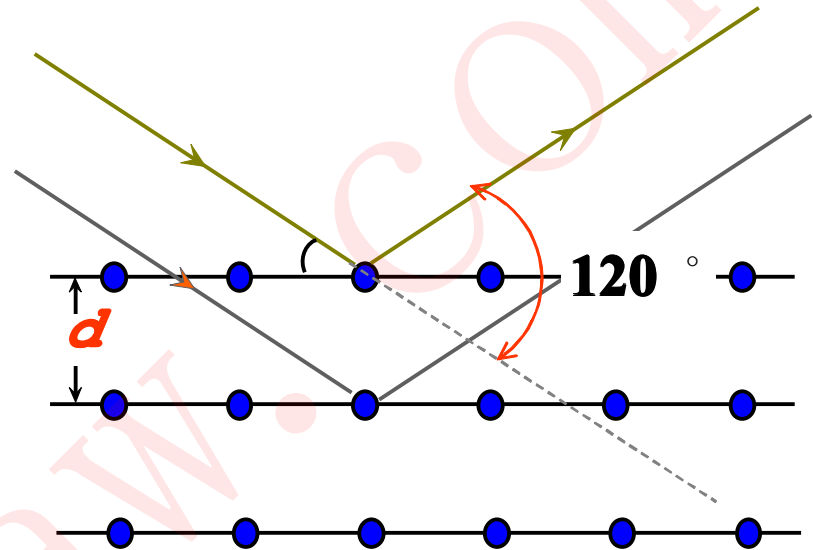
或 $\because \nu = \frac{c}{\lambda} = 2.47 \times 10^{15} (Z - 1)^2$

$$\therefore \frac{3 \times 10^8}{0.0685 \times 10^{-9}} = 2.47 \times 10^{15} (Z - 1)^2 \Rightarrow Z = 43$$

6-6 $2d \sin \theta = k\lambda$

$$\because k = 1, \theta_1 = \frac{120}{2} = 60^\circ$$

$$\therefore d = \frac{\lambda}{2 \sin \theta_1} = \frac{0.54}{2 \sin 60^\circ} = 0.31 \text{ (nm)}$$



6-7 $h\nu_0 = m_0c^2 \Rightarrow \lambda_0 = \frac{h}{m_0c} = \lambda_c$

$\varphi = 180^\circ$ 散射电子能量最小

$$\Delta\lambda = \lambda_c(1 - \cos\varphi) = 2\lambda_c \Rightarrow \lambda = 3\lambda_c \Rightarrow \nu = \frac{1}{3}\nu_0$$

$$\therefore h\nu|_{\min} = \frac{1}{3}h\nu_0 = \frac{1}{3}m_0c^2 = \frac{0.511\text{MeV}}{3} = 0.17\text{MeV}$$

$$P|_{\max} = \frac{h}{\lambda} + \frac{h}{\lambda_0} = \frac{h}{3\lambda_c} + \frac{h}{\lambda_c} = \frac{4h}{3\lambda_c} = \frac{4m_0c}{3} = 3.64 \times 10^{-22} \text{ (kg} \cdot \text{m/s)}$$